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AI FOR SOCIETY

YCD Farmer Guide

AI-Powered Agricultural Support Platform

"Bridging the digital divide for smallholder farmers"

Theme:	AI for Society — Smart Agriculture & Digital Inclusion
Document:	Phase 1 Project Report
Team:	5 Members — Software, Network, Environment, Agriculture, Microbiology
Universities:	University of Buea & University of Bamenda, Cameroon
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1 EXECUTIVE SUMMARY

Cameroon's agricultural sector sustains over 70% of the rural population, yet smallholder farmers — who produce the majority of the country's food — remain trapped in a cycle of low productivity, high post-harvest losses, and limited market access. Plant diseases alone destroy up to 40% of harvests annually, while fewer than 10% of farming communities have access to trained agronomists.

YCD Farmer Guide is a fully functional, AI-powered mobile application that places advanced agricultural intelligence directly in the hands of smallholder farmers. Built by a team of five young Cameroonians under **Youths & Contemporary Development (YCD)**, the platform combines **computer vision for crop disease identification**, a **bilingual voice-enabled AI assistant**, a **farmer-to-buyer marketplace with mobile money payments**, **expert agronomist consultations**, **community forums**, and **hyperlocal weather intelligence** — all within a single mobile application that works on basic Android smartphones in low-bandwidth areas.

This is not a concept or prototype. YCD Farmer Guide is a production-grade application with 29 frontend screens, 34 backend API controllers, 43 database models, and 8 external API integrations — already deployed and running on cloud infrastructure.



2 THE PROBLEM

Smallholder farmers in Cameroon face a convergence of interconnected challenges that perpetuate poverty and food insecurity across rural communities:

- **Crop Disease Devastation:** Diseases such as Cassava Mosaic Virus, Black Sigatoka on plantain, Cocoa Black Pod, Maize Streak Virus, and Tomato Late Blight destroy up to 40% of harvests annually. Farmers lack tools to identify what is affecting their crops and cannot access timely, reliable treatment advice.
- **Knowledge Gap:** Fewer than 10% of farming communities have access to trained agronomists or government extension services. The gap between agricultural research and the farmers who need practical guidance remains vast.
- **Climate Vulnerability:** Increasingly unpredictable weather patterns disrupt planting and harvesting cycles. Farmers make critical decisions — when to plant, when to harvest, when to apply treatments — without access to localized weather data.
- **Market Exploitation:** After a successful harvest, farmers are often forced to sell through exploitative middlemen at prices well below fair market value. They lack direct access to buyers and transparent pricing information.
- **Digital Exclusion:** Existing agricultural technology solutions are designed for literate, connected, and technologically savvy users. Farmers with limited literacy, who speak primarily French or local languages, and who operate in low-bandwidth environments are excluded from the digital revolution.
- **Isolation:** Farmers work in isolation, unable to share knowledge, learn from peers in their region, or access collective wisdom that could improve their practices.

"A farmer in Bafut who discovers brown spots on her cassava leaves has no one to ask, no tool to consult, and no way to know if her entire harvest is at risk — until it is too late."

3 OUR SOLUTION — YCD FARMER GUIDE

YCD Farmer Guide is a **comprehensive mobile application** that serves as an **all-in-one digital companion** for smallholder farmers. Rather than addressing a single pain point, the platform integrates six core capabilities into a unified experience that accompanies the farmer from field to market:



AI Disease Identification

Photograph a sick plant; within seconds the AI identifies the disease, provides treatment steps, and prevention advice — all in the farmer's language. Saves weeks of guesswork.



Voice-Enabled AI Assistant

Speak to the app in French or English and receive spoken responses. No typing, no reading required — farmers with limited literacy can access AI-powered advice instantly.



Farmer-to-Buyer Marketplace

Buy seeds, fertilizers, and farm inputs; sell produce directly to buyers at fair prices with MTN MoMo / Orange Money. Eliminates exploitative middlemen.



Expert Consultations

Book sessions with certified agronomists who verify AI diagnoses and advise farmers on correct use of fertilizers and inputs purchased on the marketplace.



Hyperlocal Weather Intelligence

7-day GPS-based forecasts tell farmers exactly when to plant, harvest, or apply treatments. Automatic severe weather alerts protect crops before damage occurs.



Community Forums

Location-based forums where farmers share tips, ask questions, and learn from peers. Breaks isolation and builds collective knowledge across regions.

Beyond these six pillars, the application includes **farm profile management** with GIS mapping (PostGIS), a **comprehensive crop database** with region-specific recommendations, **AI-generated personalized farm**

guidelines, a **digital wallet system** with escrow payments, an **educational guidance library**, a full **admin dashboard** with analytics, and **offline capabilities** with automatic data synchronization — totaling over 40 distinct features.

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TECHNOLOGY & ARCHITECTURE

YCD Farmer Guide is built on a modern, production-grade technology stack designed for reliability, scalability, and performance in low-resource environments:

Layer	Technology	Purpose
Mobile App	React Native + Expo SDK 54	Cross-platform mobile app (Android/iOS)
UI Framework	React Native Paper + i18next	Material Design UI with full FR/EN localization
Backend API	Node.js + Express.js	RESTful API server with 34 controller modules
Database	PostgreSQL + PostGIS + Sequelize	Relational DB with geospatial queries (43 models)
AI Vision	Groq API — LLaMA 4 Scout 17B	Real-time plant disease identification from photos
AI Chat	Groq API — LLaMA 3.3 70B	Context-aware agricultural chatbot
Speech-to-Text	Groq API — Whisper Large V3	Voice input transcription (FR/EN)
Text-to-Speech	Google Cloud TTS	Spoken AI responses with audio caching
Weather	Tomorrow.io + Open-Meteo	Hyperlocal forecasts with automatic failover
Maps	MapTiler + OpenStreetMap Overpass	Farm mapping and nearby market discovery
Payments	MTN MoMo + Orange Money	Mobile money integration + digital wallet + escrow
Real-time	Socket.IO	Live updates for orders, forums, products, consultations
Email	Brevo (Sendinblue)	Transactional emails: verification, alerts, notifications
Images	Cloudinary	Cloud image storage for products, profiles, certifications
Caching	Redis (Upstash) + MMKV	Server-side and client-side caching layers
Hosting	Railway Cloud	Production deployment with CI/CD
ML Backup	YOLOv8 (Python)	Offline batch disease detection pipeline

Architecture Overview

The system follows a **client-server architecture** with the React Native mobile app communicating with the Node.js backend via RESTful APIs and WebSocket connections. The backend orchestrates all AI services through the Groq API, manages the PostgreSQL database with PostGIS for geospatial operations, and handles real-time event distribution via Socket.IO. A two-tier caching strategy (Redis server-side + MMKV client-side) ensures fast response times even in low-bandwidth conditions. The offline data manager with SQLite provides full functionality when connectivity is unavailable, with automatic synchronization when the network returns.

5 COMPLETE FEATURE CATALOG

YCD Farmer Guide includes over 40 features across 29 screens. Below is the complete catalog organized by domain:

■ Authentication & Security

- Multi-role registration (Farmer, Expert, Admin) with email verification — ensures trusted users on the platform
- JWT token-based authentication with role-based access control — protects sensitive farm and financial data
- Google and Facebook OAuth social sign-in — simplifies onboarding for farmers already using social media
- Secure password reset with email-based tokens
- Admin approval workflow — ensures only verified experts can offer consultations
- Brute-force protection (auto-block after 5 failed attempts) and tiered rate limiting
- Input sanitization against XSS/injection and comprehensive audit logging with IP tracking

■ AI & Machine Learning

- AI plant disease identification via LLaMA 4 Scout vision model — farmers photograph a sick plant and get a diagnosis in seconds, saving weeks of uncertainty
- Disease database: 7 conditions across 6 crop families with specific treatments for Cameroon
- Camera capture and gallery image selection — works with any Android smartphone camera
- Multi-language AI results (auto-translation EN↔FR) — the farmer always receives results in their preferred language
- Farm-context-aware AI chatbot (LLaMA 3.3 70B) — answers any farming question using the farmer's actual crop, soil, weather, and advisory data
- Voice input via Whisper Large V3 speech-to-text (FR/EN) — farmers speak their questions instead of typing, essential for low-literacy users
- Voice output via Google Cloud TTS with audio caching — the app reads answers aloud so farmers can listen while working in the field
- Voice command system with intent detection — hands-free navigation of the entire app
- YOLOv8 offline ML pipeline — batch disease detection even without internet connectivity
- AI-generated personalized farm guidelines per soil type, region, and crop — tailored advice, not generic tips
- Smart suggestion engine combining weather + crop + market data for optimal farming decisions

■ Weather System

- Real-time weather: temperature, humidity, wind, UV, precipitation — farmers know conditions before going to the field
- 7-day hyperlocal forecast with hourly granularity — helps plan planting, spraying, and harvesting at the right time
- GPS-based automatic location detection — no manual setup needed
- Severe weather alerts with push notifications — protects crops by giving advance warning of storms and droughts
- Crop-weather impact analysis with actionable advisories — the app explains what the weather means for each specific crop
- Scheduled auto-updates: hourly weather, daily forecasts (5 AM) — fresh data every morning before fieldwork
- Dual API strategy: Tomorrow.io (primary) + Open-Meteo (fallback) — ensures weather data is always available

■ Farm Management

- Farm profiles with GPS coordinates, region, soil type, water source — the digital identity of each farm

- Multi-farm support — manage several farms per account, ideal for farmers with dispersed plots
- Crop tracking with planting dates and growth stages — never lose track of what was planted when
- Comprehensive crop database with growing guides and disease info — a farming encyclopedia in every pocket
- PostGIS-powered farm mapping with plot subdivisions — visualize farm boundaries on a real map
- Interactive MapTiler map for farm location selection — tap to set your farm's exact position
- Soil data service with cached lookups by coordinates — understand your soil before planting
- Region-specific crop recommendations with fuzzy matching — know which crops thrive in your area

■ Marketplace & E-Commerce

- Product listings with images (Cloudinary), categories, and pricing — farmers browse and buy seeds, fertilizers, and tools
- Full shopping cart: add, update, remove items with delivery fees — familiar online shopping experience
- Checkout flow with address selection (10 Cameroon regions) — nationwide delivery coverage
- MTN Mobile Money and Orange Money payment integration — the payment methods 99% of users already have
- Cash on Delivery payment option — builds trust for first-time buyers
- Digital wallet system (XAF currency) with deposit/withdraw — a secure in-app financial account
- Escrow payment system with release conditions — protects both buyers and sellers from fraud
- Order management with real-time Socket.IO status updates — track every order from purchase to delivery
- Tiered commission system: virtual 15%, on-site 20%, emergency 25% — transparent and fair
- Nearby market discovery via OpenStreetMap Overpass API — find physical markets around you

■■■ Expert Consultations

- Expert profiles with specializations, certifications, and languages — farmers choose the right expert for their problem
- Consultation booking with date, duration, type (remote/on-site) — flexible scheduling
- Real-time chat between farmer and expert with image sharing — show the expert exactly what the crop looks like
- Experts advise on correct use of fertilizers and inputs purchased on the marketplace — ensuring products are applied properly
- Voice-enabled consultation requests — speak the problem, no typing required
- Admin review workflow before expert assignment — quality assurance on every consultation
- Expert rating and review system — community-driven quality tracking
- Cost estimation with real-time commission breakdown — full price transparency

■ Community Forums

- Create and browse forum topics with categories — organized knowledge sharing
- Location-based forum discovery by GPS and radius — find discussions relevant to your area
- Region-based filtering (10 Cameroon regions) — connect with farmers facing similar conditions
- Threaded discussions with posts and replies — structured conversations for easy follow-up
- Real-time forum chat within topics — instant peer-to-peer help
- Forum membership, join/leave functionality — curate your community
- Content moderation system — ensures discussions remain helpful and respectful

■ Education & Guidance

- Built-in knowledge library: crop planning, soil management, pest control — learn anytime, even offline
- Farm-specific guidelines generated from soil, region, and crop data — advice tailored to your exact farm

- Swipeable guidelines carousel for easy browsing — intuitive touch navigation
- Drill-down navigation from categories to subsections — find specific information quickly

■ Internationalization

- Full bilingual UI: French and English across all 29 screens — no feature is language-locked
- One-tap language switching (FR↔EN) stored persistently — switch anytime without losing your place
- Backend language-aware responses via Accept-Language headers — AI and system messages adapt automatically
- Downloadable language packs for offline TTS content — listen to advice even without internet

■ Real-Time & Offline

- Socket.IO real-time updates for orders, products, forums, consultations — everything stays current
- Offline data manager with SQLite local database — full app functionality without internet
- Priority-based sync queue with retry on reconnection — no data is ever lost
- Two-tier cache: MMKV (client) + Redis (server) with stale-while-revalidate — fast access even on slow networks
- Offline status indicator with pending sync count — farmers know exactly what's waiting to sync
- Connectivity monitoring with automatic sync trigger — seamless transition between online and offline

■ Admin Dashboard

- Analytics dashboard: users, products, farms, advisories, experts — real-time platform health at a glance
- User management and approval/rejection workflow — maintain platform quality
- Expert management with certification upload review — verify credentials before experts advise farmers
- Product management for marketplace curation — ensure only quality inputs are listed
- Forum management and topic creation — guide community discussions
- E-commerce analytics: carts, orders, payments, revenue tracking — full business intelligence
- Activity monitoring and audit log viewer — security and accountability

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AI & MACHINE LEARNING PIPELINE

Artificial intelligence is at the core of YCD Farmer Guide. The platform leverages three distinct AI models through the Groq API, each serving a specific role in the farmer's journey:

Model	Provider	Role	Details
LLaMA 4 Scout 17B 16E Instruct	Groq (Meta)	Disease Identification	Vision model analyzes plant photos. Returns JSON: disease name, confidence score, severity (mild/moderate/severe).
LLaMA 3.3 70B Versatile	Groq (Meta)	AI Chatbot & Translation	Farm-context-aware chatbot that loads the farmer's crops, soil type, weather, and recent advisories into its context.
Whisper Large V3	Groq (OpenAI)	Speech-to-Text	Transcribes voice recordings (M4A/WAV) to text in French and English. Enables farmers with limited literacy.
Google Cloud TTS	Google	Text-to-Speech	Converts AI responses to spoken audio (MP3). Supports male/female/neutral voices in EN/FR. Cached audio for offline use.
YOLOv8 (Backup)	Ultralytics	Batch Detection	Python-based offline detection pipeline using YOLOv8 for environments without reliable internet. Serves as a backup for disease identification.

Disease Detection Workflow

1. The farmer opens the Disease Detection screen and either captures a photo with the device camera or selects one from the gallery.
2. The image is encoded as base64 and sent to the backend API, which forwards it to the Groq Vision API with a specialized prompt for African crop pathology.
3. The LLaMA 4 Scout model analyzes the image and returns a structured JSON response including the disease name, confidence score (0-1), severity level, specific treatment steps for Cameroon, and prevention advice.
4. The backend enriches this result with curated data from the local disease database (7 diseases with detailed treatments). If the farmer's language is French, the result is automatically translated by the LLM.
5. The farmer receives the complete diagnosis on screen. They can then consult the AI chatbot for follow-up questions or book an expert consultation for human verification.

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INNOVATION & DIFFERENTIATION

YCD Farmer Guide is distinguished by several innovations that set it apart from existing agricultural technology solutions:

- **Voice-First Design for Inclusivity:** While most agritech apps assume literate, tech-savvy users, YCD Farmer Guide was designed voice-first. Farmers can speak to the app in French or English and receive spoken responses — no typing, no reading required. This makes advanced AI accessible to users with limited literacy, a population entirely excluded by existing solutions.
- **All-in-One Ecosystem:** Rather than offering a single feature (disease detection OR marketplace OR weather), YCD Farmer Guide integrates the entire farmer journey — from crop health monitoring to expert consultations to selling produce — in one app. This reduces the burden of downloading and learning multiple tools.
- **Human-AI Collaboration:** The app does not position AI as a replacement for human expertise. Instead, farmers can verify AI-generated diagnoses by booking consultations with certified agronomists directly within the same app. Experts also advise farmers on the correct use of fertilizers and inputs purchased through the marketplace, ensuring products are applied safely and effectively. This builds trust and ensures accuracy.
- **Offline-First Architecture:** SQLite local database with priority-based sync queue means the app works without internet. When connectivity returns, data automatically synchronizes. A stale-while-revalidate cache pattern ensures farmers always see data even in network dead zones.
- **Local Payment Integration:** Instead of international payment gateways, the app integrates MTN Mobile Money and Orange Money — the payment methods actually used by 99% of the target population. An escrow system protects both buyers and sellers.
- **Geospatial Intelligence:** PostGIS-powered farm mapping, OpenStreetMap market discovery, GPS-based weather, and location-aware community forums — all leveraging geolocation to provide contextually relevant information.

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SOCIAL IMPACT & SDG ALIGNMENT

YCD Farmer Guide directly contributes to multiple United Nations Sustainable Development Goals, making measurable impact in the communities it serves:

SDG	Goal	Contribution
SDG 1	No Poverty	Increases farmer income through direct market access, fair pricing, and elimination of exploitative middlemen.
SDG 2	Zero Hunger	Reduces crop losses through AI disease identification, weather alerts, and expert agricultural guidance — improving food security.
SDG 4	Quality Education	Provides an educational platform with farming knowledge base, AI-powered guidance, and peer-to-peer learning via community groups.
SDG 8	Decent Work	Creates economic opportunities: experts earn from consultations, farmers sell at fair prices, and digital jobs emerge around the platform.
SDG 9	Industry & Innovation	Applies cutting-edge AI (LLaMA 4, Whisper) to solve traditional agricultural challenges — technology innovation for social good.
SDG 10	Reduced Inequality	Voice interface and bilingual design ensure digital inclusion for low-literacy and marginalized farming communities.
SDG 12	Responsible Consumption and Production	AI-guided treatments reduce pesticide overuse; marketplace reduces food waste by connecting supply to demand.
SDG 13	Climate Action	Weather intelligence and crop-weather impact analysis help farmers adapt to climate change and extreme weather events.
SDG 17	Partnerships	Built by a multidisciplinary team under a social enterprise, engaging universities, communities, and technology partners.

Measurable Impact Targets

- 50,000+ smallholder farmers reached across all 10 regions of Cameroon within 2 years
- 200 rural digital ambassadors trained to help farmers adopt the technology
- 30% reduction in crop losses for active users through AI-guided disease management
- Direct market access for farmers, reducing dependency on middlemen by 50%
- 100+ certified agronomists onboarded to provide expert consultations

IMPLEMENTATION STATUS & SCALABILITY

YCD Farmer Guide is **not a concept or mockup**. The application is fully built, tested, and deployed:

Metric	Status
Frontend Screens	29 screens (18 main + 11 admin) — fully implemented
Backend Controllers	34 API controller modules — all operational
Database Models	43 Sequelize models with PostGIS — production-ready
Database Migrations	Full migration suite for automated deployment
AI Integration	3 Groq models + Google TTS — live and tested
Payment System	MTN MoMo + Orange Money + Wallet + Escrow — integrated
Real-Time	Socket.IO with 10+ event types — operational
Deployment	Railway Cloud (backend) + EAS Build (mobile) — deployed
Version Control	Git with 2 remote repositories on GitHub
Code Volume	34 controllers, 34 services, 32 components, 43 models, 35 routes

Scalability Roadmap

The platform is architected for progressive expansion across multiple dimensions:

- **Voice in Local Languages:** The voice interface currently supports French and English. The next phase will extend speech-to-text and text-to-speech to major Cameroonian languages (Pidgin English, Ewondo, Fulfulde, Duala, Bamiléké) — enabling the 60%+ of rural farmers who communicate primarily in local languages to use the app in their mother tongue.
- **Video & Audio Calls with Experts:** Expert consultations currently use real-time text chat with image sharing. The planned upgrade introduces in-app video and audio calling, allowing farmers to show their fields live and receive immediate verbal guidance from agronomists — critical for urgent crop emergencies.
- **Pan-African Expansion:** The crop disease database and regional intelligence can be populated with data from any African country. The mobile money integration can be adapted to M-Pesa, Airtel Money, and other regional providers.
- **Infrastructure Scaling:** Railway Cloud auto-scales based on demand. PostgreSQL supports horizontal read replicas. The architecture is designed to serve 100,000+ concurrent users without redesign.

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THE TEAM

YCD Farmer Guide is built by a **multidisciplinary team of five Cameroonians** working under **Youths & Contemporary Development (YCD)**, a social enterprise based in Yaoundé dedicated to empowering youth through innovation and sustainable development. The team spans five disciplines across multiple institutions:

Ikome Johnson (Team Lead)

Team Lead — Systems & Connectivity
Network Engineering (Level 500) — University of Buea

Project coordination, systems integration, connectivity architecture, and deployment strategy.

Mkounga Tatchum Laurent Joël

Technical Lead — Full-Stack & AI
Software Engineering — University of Buea

System architecture, full-stack development (React Native, Node.js), AI/ML integration (Groq, LLaMA, Whisper), database design, and platform engineering.

Ndum Albert Teghen

Mentor — Environmentalist
Environmental Science — —

Strategic mentorship, environmental sustainability guidance, ecological impact assessment, and project advisory.

Ndinguru Hope

Agriculture Specialist
Agriculture — ISAGO (Institut Supérieur Agricole d'Obala)

Agricultural domain expertise, crop science validation, farmer engagement strategy, and field-level advisory content.

Tata Stanley Lem-Mola

Crop Health & Biological Systems
Microbiology — University of Bamenda

Disease detection domain knowledge, biological systems validation, crop health data curation, and scientific accuracy review.

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"Technology that speaks the language of farmers and solves real problems in the field."